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Toughened self-assembled monolayers for durable perovskite solar cells

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Hole-selective self-assembled monolayers (SAMs)[1,2] have played a key role in driving the certified power conversion efficiency (PCE) of inverted perovskite solar cells[3–5] to 26.7% (ref. 6). However, their instability often compromises the operational performance of devices, strongly hindering their practical applications[7,8]. Here we employ a cross-linkable co-SAM to enhance the conformational stability of hole-selective SAMs against external stresses, while suppressing the formation of defects and voids in SAM during self-assembly. The azide-containing SAM can be thermally activated to form a cross-linked and densely assembled co-SAM with a thermally stable conformation and preferred orientation. This effectively minimizes substrate surface exposure caused by wiggling of loose SAMs under thermal stress, preventing perovskite decomposition. This enables a certified PCE of 26.92% to be achieved for the best-performing cell, which also possesses excellent thermal stability with negligible decay under maximum-power-point tracking at 85 °C for 1,000 h. It also retains >98% of initial PCE after 700 repetitive thermal cycles between −40 °C and 85 °C, representing the state of the art of the field. This work offers an in-depth understanding of SAM degradation mechanisms to guide the design of a more robust buried interface for SAM-based devices adopting high-roughness substrates to realize highly efficient and durable perovskite solar cells.

Inverted perovskite solar cells (PSCs) have garnered considerable attention due to their progressively enhanced operational stability and compatibility with tandem devices[3–5,9]. The research on inverted PSCs has recently gained impetus with the advent of carbazole-derived hole-selective SAMs, which significantly boost device power conversion efficiency[10]. Although these SAMs enable efficient hole extraction and effective surface passivation[11–16], their molecular instability[7,8] and uneven coverage on the substrate[17] undermine device durability. Although this issue can be alleviated by adopting a co-SAM or a metal-oxide–SAM composite[18,19], the heterogeneous SAM–perovskite interface is still highly vulnerable. Moreover, micelle formation in the SAM precursor solution[20] and desorption of SAM molecules during subsequent processing[17] compromise SAM quality. It is therefore imperative to significantly enhance SAM stability through molecular engineering.

State-of-the-art SAM molecules typically possess a phosphonic acid anchor and conjugated head group, bridged by an alkyl or aryl linker[21]. We have demonstrated in a past work via CbzNaph[22] that SAM stability can be enhanced by extending the π-conjugation of the head groups to strengthen intermolecular π–π interactions. Unlike rigid conjugated head groups, the flexible alkyl linkers offer no resistance to external stimuli. Although the alkyl linkers usually possess comparable or longer lengths than that of the conjugated head groups, they occupy considerably less volume. Consequently, the monolayer formed can be viewed as a compact top shell composed of π-interacting conjugated head groups covering atop the loosely arranged, skinny alkyl linkers. The vast free volume among the linkers allows the flexible and weakly interacting alkyl chains to deform easily under external stimuli. This significantly deteriorates PSC operational stability, posing a critical challenge to the field.

Herein we report the employment of a cross-linkable co-SAM as a toughened hole-selective layer (HSL). We develop an azide-containing cross-linkable guest SAM (JJ24) capable of forming robust covalent bonds with the nearby alkyl linkers from the host SAM (CbzNaph). The cross-linked co-SAM (CbzNaph:JJ24) is highly effective at suppressing SAM conformational changes under external stresses to mitigate the direct contact between perovskite and transparent conductive oxide (TCO) for preventing perovskite decomposition. Moreover, the introduction of JJ24 also helps disassemble CbzNaph micelles in the mixed-CbzNaph:JJ24 precursor solution to improve solute dispersion. The cross-linked co-SAM not only improves substrate coverage but also maintains a preferred orientation vertical to the substrate as corroborated both experimentally and theoretically. Furthermore, the collective molecular dipole of co-SAM is enhanced to down-shift the TCO work function, benefiting interfacial hole extraction. Consequently, the champion inverted PSC adopting cross-linked co-SAM achieves a certified PCE of 26.92%. The encapsulated device shows negligible decay of its initial PCE after maximum-power-point (MPP) tracking at 85 °C for

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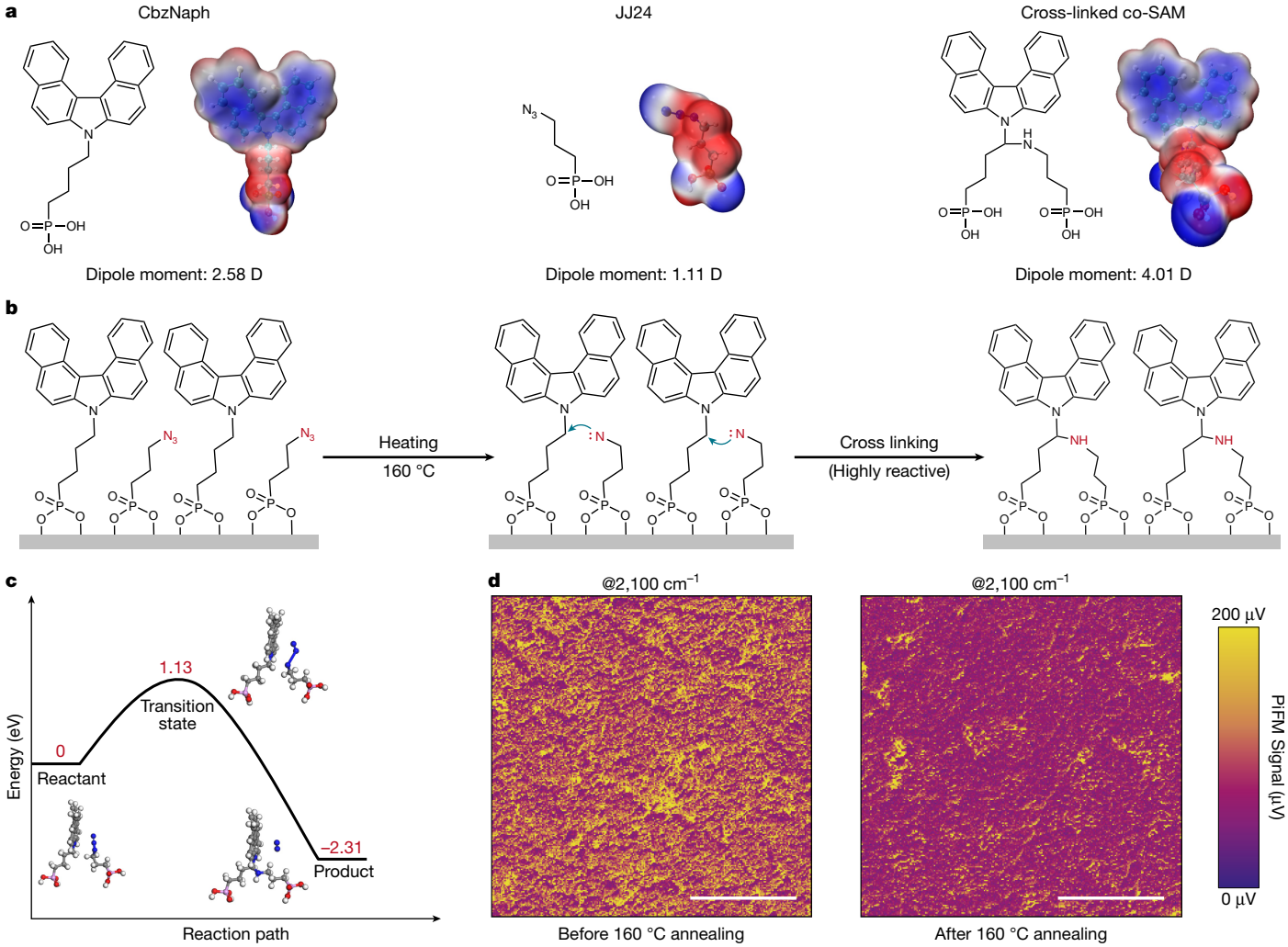

Fig. 1 | Molecular design and cross-linking mechanism of toughened SAMs. **a**, Chemical structures, electrostatic potentials and molecular dipoles of CbzNaph, JJ24 and the proposed cross-linked target product. The resulting –NH groups are positioned near the electron-rich 7*H*-dibenzo[*c*,*g*]carbazole skeleton, which enables the cross-linked target molecules to have significantly improved molecular dipoles to regulate the work function of substrate more effectively. **b**, The proposed cross-linking mechanism between CbzNaph and JJ24. **c**, The DFT calculation for the reaction path and Gibbs free energy between CbzNaph and JJ24. **d**, The PiFM images are acquired at 2,100 cm^{-1} for the substrates coated with CbzNaph and JJ24 before and after thermal annealing at 160 °C for 15 min. Scale bars, 1 μm.

1,000 h under continuous 1-sun illumination in ambient atmosphere with a high relative humidity of 70–80% (ISOS-L-2 protocol). Moreover, it can retain over 98% of its initial PCE after 700 repetitive thermal cycles between −40 °C and 85 °C, representing the forefront results of the field.

Molecular design and working mechanism

The cross-linkable SAM molecule JJ24 has an azide head group, a propyl linker and a phosphonic acid anchor (Fig. 1a). Although organic azides can present safety concerns, JJ24 complies with safety standards (the rule of six)[23] and remains reasonably stable when stored over the long term (Supplementary Fig. 1). Thermal activation converts the azide group into nitrene to enable JJ24 to covalently cross-link with the alkyl linkers of host CbzNaph (Fig. 1b)[24]. The phosphonic acid on JJ24 allows for its co-deposition with common phosphonic-acid-based SAM molecules due to their similar pK_a values[1,25]. Its linear propyl linker is designed to be shorter than the butyl linker of the host CbzNaph, so it not only can fill the voids but also enhance the van der Waals interactions between SAM molecules. This also ensures the desired topological cross-linking of nitrene with the sp^3 carbon atom adjacent to the CbzNaph head group. Thermogravimetric analysis reveals the onset of azide decomposition at 148 °C, which is the temperature azide releases nitrogen to form nitrene (Supplementary Fig. 2)[26,27]. Density functional theory (DFT) calculations confirm a favoured Gibbs free energy of −2.31 eV with a modest energy barrier (1.13 eV) (Fig. 1c). Importantly, the cross-linked co-SAM exhibits an enhanced dipole moment (4.01 D versus 2.58 D for pristine CbzNaph) (Fig. 1a), which can effectively modulate the substrate work function.

CbzNaph:JJ24 co-SAM is processed in a 2:1 molar ratio to achieve a balance between cross-linking density and perovskite wettability. A higher JJ24 concentration helps improve SAM coverage but deteriorates surface wettability[28,29] (Supplementary Fig. 4) and perovskite crystallization. High-resolution X-ray photoelectron spectroscopy (HR-XPS) measurements confirm that the optimal co-SAM density is at a ratio of 2:1 (Supplementary Figs. 5 and 6, and Supplementary Table 1), consistent with contact-angle measurements.

Formation of cross-linked CbzNaph:JJ24 co-SAM and evidence

Self-assembled monolayer formation is strongly dependent on the dispersion of SAM molecules in solution and the hydroxyl group density on

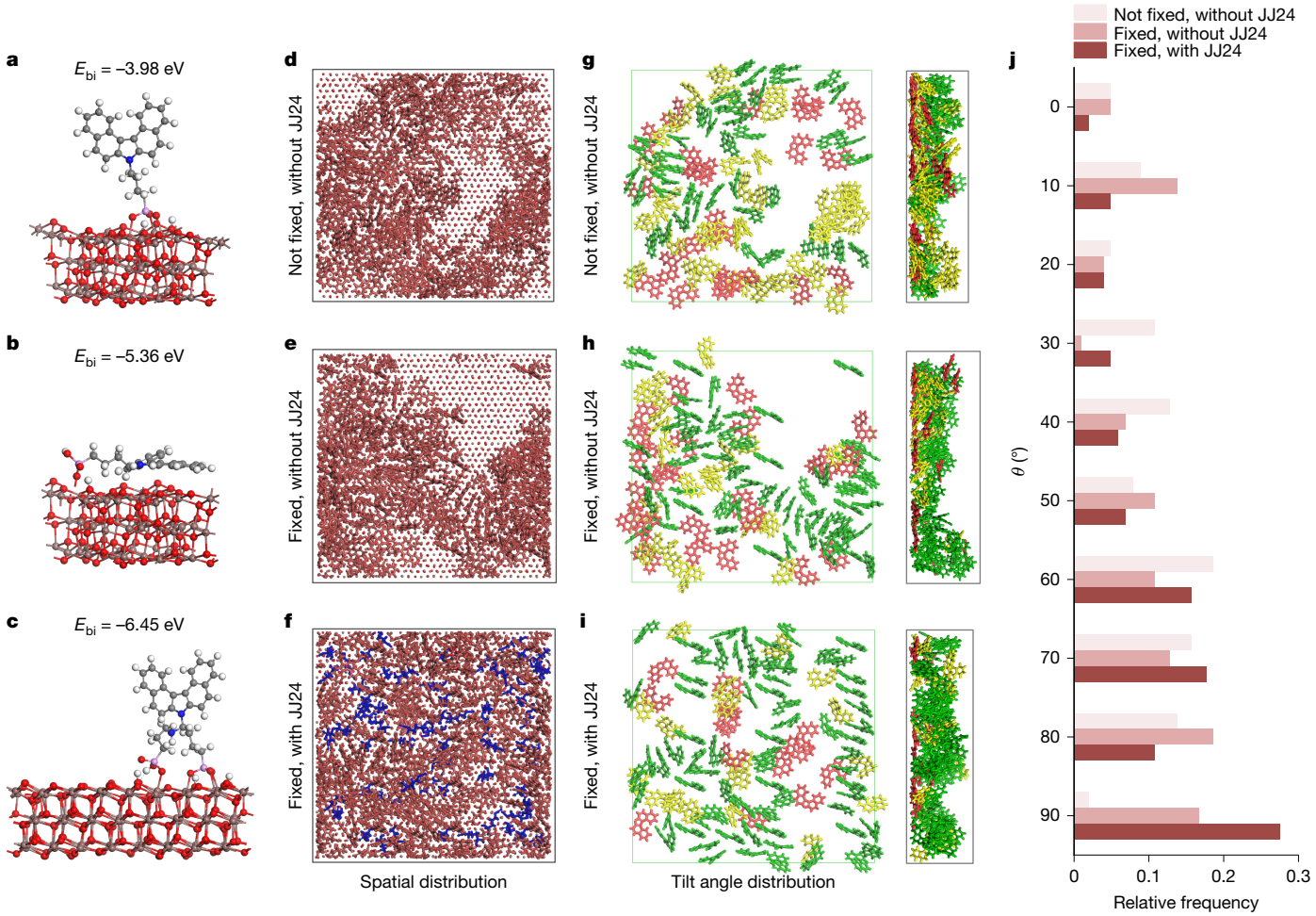

Fig. 2 | Molecular dynamics simulations of host–guest SAMs. a–c, The DFT-calculated binding energies of CbzNaph in edge-on binding orientation (**a**), face-on binding orientation (**b**) and after cross-linking with JJ24 (**c**), together with semi-quantitative HR-XPS analysis based on the P/In ratio (0.024 for CbzNaph versus 0.083 for CbzNaph:JJ24 (2:1)), confirm a marked increase in the surface assembly density of CbzNaph:JJ24 on ITO substrates. By integrating the SAM density with DFT-derived binding energies, the cross-linked CbzNaph:JJ24 (2:1) system exhibits a [0.083 × 2/3 × (−3.98) + 0.083 × 1/3 × (−2.47)]/[0.024 × (−3.98)] = 3.02-fold enhancement in substrate binding energy per unit area compared to CbzNaph alone. **d–i**, The molecular-dynamics-simulated spatial distributions of CbzNaph (**d–f**), and the tilt angle distribution of CbzNaph (**g–i**) on the ITO substrates without and with JJ24. The colours of the molecules indicate different ranges of tilt angles with respect to the ITO substrate. **j**, The histogram of the tilt angle (with respect to the substrate) of CbzNaph without and with JJ24. Fixed means that a small portion of CbzNaph molecules are restricted with a vertical orientation with respect to the substrate and unable to move during simulations; CbzNaph molecules prefer to aggregate on the ITO substrate regardless of whether a small portion of CbzNaph molecules is restricted or not.

the TCO surface. Although the commonly observed micelle formation in alcoholic solvents can be dispersed with polar additives[20], these additives tend to deplete TCO surface hydroxyls. The introduction of JJ24 can suppress micelle formation via hydrogen bonding (phosphonic acids) and van der Waals interactions (alkyl linkers), as confirmed by dynamic light scattering (Supplementary Fig. 7). The suppressed SAM aggregation in solution facilitates the subsequent denser assembly on the TCO substrate without using polar additives. To conduct in-depth characterizations of SAM on TCO, indium-tin oxide (ITO) substrate is used instead of fluorine-doped tin oxide (FTO) due to its lower surface roughness.

Cross-linking between CbzNaph and JJ24 is triggered by annealing the co-SAM at 160 °C to activate the azide group (Supplementary Fig. 8). The azide characteristic Fourier-transform infrared peak (~2,100 cm^{-1}) is monitored through photo-induced force microscopy (PiFM) to ensure reaction completion (Fig. 1d). The spatial mapping of the 7*H*-dibenzo[*c*,*g*]carbazole C–N stretch (~1,250 cm^{-1}) further reveals enhanced CbzNaph density with the introduced JJ24 (Supplementary Fig. 9). High-resolution XPS further confirms cross-linking through the disappearance of the high-binding-energy nitrogen signal from azides after annealing at 160 °C (ref. 30), whereas this signal persists at 100 °C (the typical annealing temperature for SAM-TCO; Supplementary Figs. 10 and 11). Direct ^{1}H NMR evidence of the cross-linked product is obtained by desorbing the co-SAMs ultrasonically from ITO in dimethyl sulfoxide-d_6 (DMSO-d_6). The characteristic up-field-shifted 7*H*-dibenzo[*c*,*g*]carbazole *N*-CH_x protons (4.65–4.72 ppm) are caused by electron donation from the newly formed –NH group (Supplementary Fig. 12). This shielding effect also induces up-field shifting of all aromatic proton signals. By combining the energetics of different reaction paths calculated from DFT for nitrene and CbzNaph (Supplementary Fig. 13), the signal of the cross-linked target molecule can be identified to validate that nitrenes on JJ24 undergo in situ cross-linking with CbzNaph on the ITO substrate.

Ultraviolet photoelectron spectrometry reveals that the substrate treated with cross-linked co-SAM has a deeper work function than that treated with pristine CbzNaph (Supplementary Fig. 14). Combined with experimental evidence and molecular dynamics simulations demonstrating that the introduction of JJ24 promotes denser assembly of CbzNaph (Supplementary Table 1 and Fig. 2d–f), as well as the enhanced

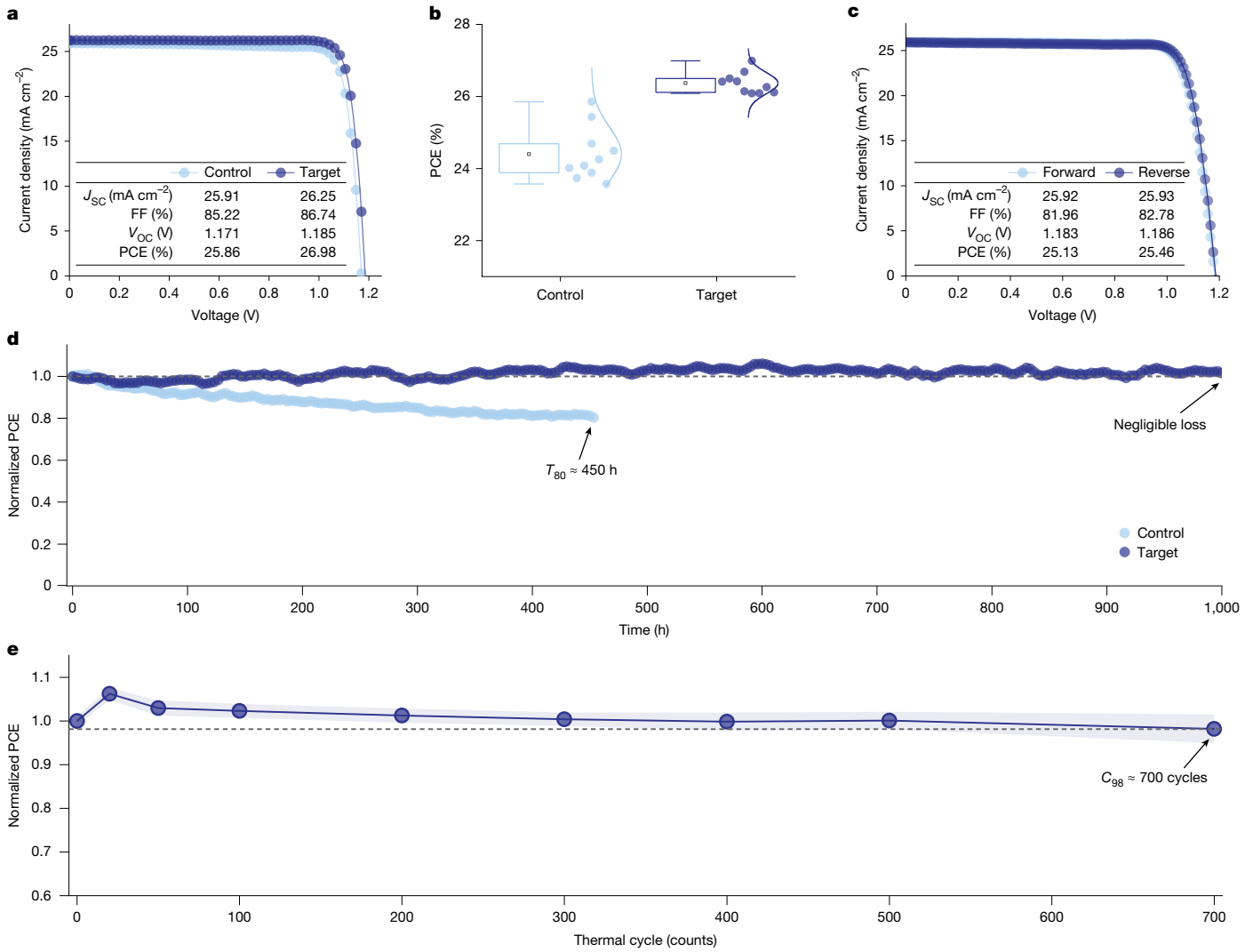

Fig. 3 | Photovoltaic performance of PSCs. a, Current density-voltage (*J*–*V*) characteristics of the control and target champion PSC. **b**, Box plots of the PCE of the control and target PSCs (10 devices). **c**, *J*–*V* curves of the champion 1 cm^2 PSC based on cross-linked SAM-HSL. **d**, Long-term operational stability of the encapsulated PSCs at MPP tracking under continuous 1-sun equivalent illumination at 85 °C in ambient air with a relative humidity of 70–80%, complied with the ISOS-L-2 protocol. The initial efficiencies of the control and target PSCs are 23.69% and 24.54%, respectively. **e**, Efficiency tracking of encapsulated target cells subjected to repetitive thermal cycling between −40 °C and 85 °C. The initial efficiency of the encapsulated PSC is 24.58%.

molecular dipole moment after cross-linking, these results indicate that favourable work function regulation and interfacial energetics alignment can be achieved through synergistically enhanced coverage and dipole moment of the cross-linked co-SAM.

The structural stability of pristine CbzNaph and cross-linked co-SAM is further investigated by XPS to evaluate their resistance and conformational stability over *N*,*N*-dimethylformamide (DMF)—a common solvent used for perovskite processing (Supplementary Figs. 15 and 16). The phosphorus atomic ratio of pristine CbzNaph SAM is obviously reduced after DMF washing, implying the desorption of more weakly bonded SAM molecules from the substrate. Conversely, only a slightly reduced phosphorus atomic ratio is found in the cross-linked co-SAM, demonstrating its exceptional solvent resistance (Supplementary Table 2).

Theoretical and experimental studies into SAM microscopic structure

Based on DFT calculations and molecular dynamics simulations, we systematically investigate how the introduced JJ24 influences the assembly of CbzNaph on ITO. The DFT calculations suggest that CbzNaph molecules tend to lie down in a face-on manner on ITO, attributable to its flexible alkyl chain and lower binding energy to ITO (that is, −3.98 and −5.36 eV for edge-on and face-on binding orientations, respectively) (Fig. 2a,b). By contrast, the cross-linked SAMs exhibit a much stronger binding energy for the edge-on orientation (−6.45 eV), rendering this orientation energetically favourable (Fig. 2c). Molecular dynamics simulations help visualize this transition from flat-lying to upright molecular arrangements upon cross-linking (Supplementary Fig. 17). Beyond optimizing molecular orientation, the introduction of JJ24 provides additional anchoring points to significantly enhance substrate adhesion. Combined with the P/In ratio calculated by HR-XPS, we estimate that the binding energy of the cross-linked SAM increases 3.02-fold compared with that of pristine SAM, which significantly improves its resistance to physical stresses and polar solvents.

Furthermore, molecular dynamics simulations show pristine CbzNaph molecules tend to form incomplete SAM coverage regardless of their initial anchoring conditions (Fig. 2d–f). This allows the direct contact between perovskite and ITO to increase charge recombination and perovskite degradation. Encouragingly, the introduction of JJ24 promotes uniform distribution of SAM molecules, as reflected by the increased

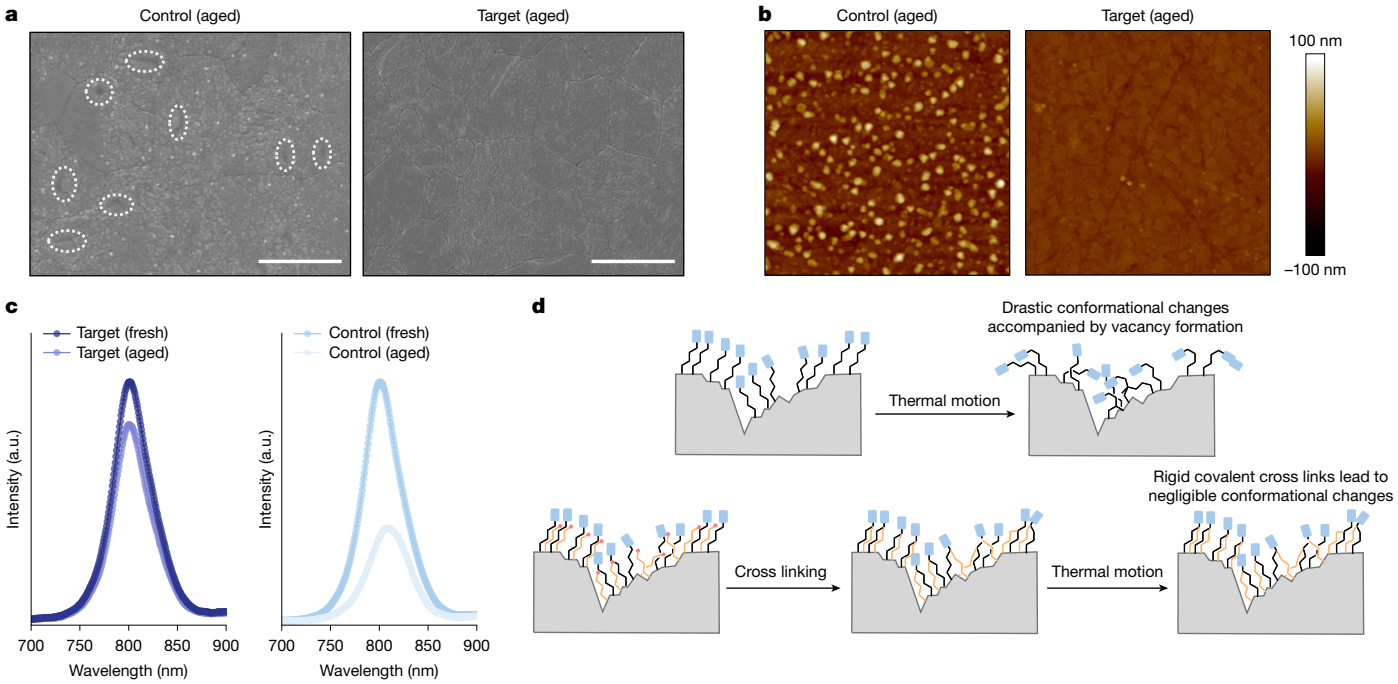

Fig. 4 | Mechanistic investigations of toughened SAMs improving the stability of PSCs. a, Top-view SEM images of the bottom surface of aged perovskite film (peeling from ITO-glass substrates with an epoxy encapsulant) deposited on CbzNaph (control) and cross-linked CbzNaph:JJ24 (target) with decomposed regions marked by dashed ovals. Scale bars, 1 µm. **b**, The AFM topographical images of CbzNaph- or cross-linked CbzNaph:JJ24-treated ITO substrates, which are prepared from the aged perovskite films with dynamic DMF washing. **c**, Photoluminescence of perovskite films based on CbzNaph and cross-linked CbzNaph:JJ24 before and after ageing. The direction of incident light comes from the glass side. **d**, Schematic diagram of the random thermal swing of SAM molecules leading to substrate exposure and subsequent perovskite decomposition.

molecular tilt angles of cross-linked SAMs (Fig. 2g–j). X-ray reflectivity measurements show monolayer thicknesses evolving from 0.72 nm (~60° tilt) for pristine CbzNaph to 1.32 nm (~24° tilt) for cross-linked co-SAM (Supplementary Fig. 18), validating JJ24's ability in promoting more vertical molecular arrangement. It provides a molecular-scale topological template to enhance the efficiency of cross-linking between CbzNaph and JJ24. This phenomenon is supported by the observations from proton NMR that shows selective cross-linking occurred at specific molecular sites (Supplementary Fig. 12).

Solar cell performance and long-term stability

We fabricated inverted PSCs to evaluate the impact of a SAM on the performance of devices with the configuration: FTO/SAM/perovskite/ethylenediammonium diiodide ($EDAI_2$)/C_{60}/SnO_x/Ag, where the SAM is either pristine (control) or a cross-linked co-SAM (target). Systematic optimization of the co-SAM reveals that the one based on the 2:1 ratio yields the best performance, whereas that of 1:1 ratio exhibits a reduced short-circuit current density (J_{SC}) and fill factor due to impaired perovskite wettability (Supplementary Figs. 19 and 20, and Supplementary Table 3). The champion device adopting the target co-SAM achieves a PCE of 26.98%, surpassing that of the control (25.86%) (Fig. 3a and Supplementary Fig. 21). The integrated J_{SC} (25.43 mA cm^{-2} for target and 24.96 mA cm^{-2} for control) from the external quantum efficiency (EQE) spectrum matches well with those from J–V sweeps (Supplementary Fig. 23). The encapsulated target device is examined by a third-party organization (National Photovoltaic Industry Measurement and Testing Center, NPVM) to afford a certified PCE of 26.92% with a V_{OC} of 1.185 V, a J_{SC} of 26.18 mA cm^{-2} and a fill factor of 86.75% (Supplementary Fig. 24). Notably, target devices exhibit narrower PCE distributions (Fig. 3b and Supplementary Fig. 20), indicating superior reproducibility due to improved coverage and interfacial energetic alignment from cross-linked co-SAM.

We perform comprehensive characterization of the properties of control and target devices to more deeply understand the impact of target SAMs on device performance. Electrochemical impedance spectrometry reveals that the PSCs with cross-linked SAMs have a lower series resistance (R_S), decreasing from 72.81 Ω (control) to 49.71 Ω, and a higher recombination resistance (R_{Rec}), increasing from 42 kΩ to over 53 kΩ (Supplementary Fig. 25), suggesting improved charge extraction and suppressed interfacial recombination. Dark J–V measurements further support this observation, showing reduced leakage current in the target device (Supplementary Fig. 26). This improvement may stem from the more uniform and denser SAM coverage when JJ24 is introduced, which effectively suppresses current leakage to boost J_{SC} (ref. 31). Mott–Schottky analysis indicates a higher built-in potential (V_{bi}) for target devices (0.95 V versus 0.87 V; Supplementary Fig. 27) due to the enhanced dipole moment of cross-linked SAMs (Fig. 1a), which strengthens the interfacial charge selectivity and mitigates the V_{OC} loss[32–35]. Trap density-of-states analysis reveals suppressed deep-state traps (0.35–0.45 eV) in target devices (Supplementary Fig. 28). Photoluminescence quantum yield (PLQY) measurements further validate the reduced non-radiative losses, with PLQY increasing from 1.2% (pristine SAM) to 6.2% (cross-linked co-SAM) and quasi-Fermi level splitting rising from 1.131 eV to 1.167 eV (Supplementary Fig. 29). The improved PLQY and quasi-Fermi level splitting clearly show the effectiveness of cross-linked co-SAM in reducing interfacial defects and non-radiative recombination in the perovskite layer. Finally, light-intensity-dependent V_{OC} measurements reveal a lower ideality factor (Supplementary Fig. 30), confirming the suppressed trap-assisted recombination[36] within the PSCs.

The cross-linked co-SAM also shows promise in scalable fabrication, where a 1 cm^2 device achieves a PCE of 25.46% (Fig. 3c), demonstrating the advantage of better substrate coverage on large-area devices. Moreover, the cross-linking of JJ24 can be generally applied to SAMs with different functional head groups for PSC fabrication (Extended

Data Fig. 1). When JJ24 is cross-linked with 4PACz, the PCE increases from 25.30% to 25.81%. Similarly, its cross-linking with CbzPh improves the PCE from 25.37% to 26.40%, and 25.47% to 26.80% while cross-linking with CbzBT (Extended Data Table 1)[37]. These results demonstrate the advantage of applying JJ24 for enhancing PSC performance.

Testing the stability of the SAM-derived device under the ISOS-L-2 protocol (85 °C, 70–80% relative humidity, 1 sun illumination) reveals exceptional operational stability for the target device, showing negligible PCE loss after 1,000-h MPP tracking; the control device, by contrast, degraded rapidly (~20%) within 450 h (Fig. 3d). Similar stability enhancement is also observed for its 1-cm^2 device and other devices based on different cross-linked systems, including 4PACz:JJ24, CbzPh:JJ24 and CbzBT:JJ24 (Extended Data Fig. 2 and Supplementary Fig. 31). Furthermore, the PCE evolution of the target device is tested with repetitive thermal cycling between −40 °C and 85 °C in dark under ambient atmosphere (ISOS-T-2 protocol) (Fig. 3e). Impressively, the encapsulated target device can retain 98.2% of its initial PCE after 700 thermal cycles ($C_{98} \approx 700$ cycles)

Cross-linked co-SAMs help stabilize buried interface during ageing

Conventional SAMs often suffer from insufficient substrate coverage due to the random motion of flexible alkyl linkers under external stresses. To assess the impact of cross-linked co-SAMs on the buried interface of aged perovskite films and devices, we first examine the evolution of interfacial morphology. Perovskite films are prepared on the less rough ITO substrates with pristine CbzNaph and a cross-linked co-SAM as the HSL. A cross-sectional scanning electron microscopy (SEM) image reveals significant perovskite decomposition in the control sample after photothermal ageing (continuous illumination at 85 °C for 100 h), whereas the target film maintains well-defined crystalline domains (Extended Data Fig. 3). Further analysis of peeled perovskite films (before and after ageing) via SEM and X-ray diffraction (XRD) shows intensified lead iodide (PbI_2) signals and decomposed regions at the buried interface of the control one (Fig. 4a and Extended Data Fig. 4)[15]. Furthermore, atomic force microscopy (AFM) scan of DMF-washed control and target films show comparable initial morphology (Supplementary Fig. 32), but the aged control sample exhibits abundant particulate residues, whereas the target sample remains largely unchanged (Fig. 4b). These particles are identified as PbI_2/Pb^0 by energy-dispersive X-ray spectroscopy (Supplementary Table 4 and Supplementary Fig. 33).

We attribute PbI_2 formation in control sample to direct contact of perovskite with ITO (ref. 38), resulting from incomplete SAM coverage. By contrast, the cross-linked co-SAM has higher molecular rigidity and denser spatial distribution, which can minimize substrate exposure. This hypothesis aligns with the observation from photoluminescence and time-resolved photoluminescence studies: the control films exhibit significantly quenched photoluminescence intensity and faster lifetime decay than target films (Fig. 4c, and Supplementary Figs. 34 and 35), indicating cross-linked co-SAM improves the quality of perovskite absorber deposited atop. It inhibits interfacial non-radiative recombination through minimized contact between perovskite and TCO, overcoming the hurdle that significantly limits device performance and stability[38,39]. This also validates the instability of SAM-based devices is caused by the disrupted SAMs during thermal ageing, leading to exposed substrate surface (Fig. 4d) that accelerates perovskite decomposition[5,40,41]. This is a general challenge for SAM-based inverted PSCs and other types of electronics.

Discussion

We report a previously undescribed cross-linked co-SAM strategy that improves both the dispersion of SAM molecules and the conformation stability of denser assemblies on TCO; this approach improves device performance and stability under stringent external stresses. Inverted PSCs adopting such toughened hole-selecting layers exhibit an excellent certified PCE of 26.92%. More importantly, the encapsulated device shows negligible degradation of its initial efficiency after being tracked at the MPP for 1,000 h at 85 °C with 70–80% relative humidity. Furthermore, it can also retain over 98% of its initial PCE after 700 repetitive thermal cycles between −40 °C and 85 °C. This represents one of the best-reported results for inverted PSCs with combined high efficiency and operational stability under accelerated ageing. This work offers significant insights for developing high-performance and stable SAM-based inverted PSCs to accelerate their potential commercialization. It also provides a solution for other SAM-based devices adopting high-roughness substrates for broader applications.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-025-09509-7.

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Methods

Materials

CbzNaph (ref. 22), 4PACz (ref. 13), CbzPh (ref. 22) and CbzBT (ref. 37) were synthesized according to the literature. Other materials were used as received without further purification. Organic halide salts, including methylammonium iodide (MAI), formamidinium iodide (FAI), $EDAI_2$ and methylammonium chloride (MACl) were purchased from Xi'an Yuri Solar. Caesium iodide (CsI) and bathocuproine were purchased from Aladdin. Lead iodide (99.99%) was purchased from TCI. Anhydrous solvents including DMF (99.8%), DMSO (99.9%), 2-propanol (IPA, 99.5%), chlorobenzene (99.8%) and methanol (99.8%) were purchased from Sigma-Aldrich; C_{60} was purchased from Advanced Election Technology; and the TCO glass of the ITO and FTO substrates was purchased from the Shaanxi Fangdecheng Construction Engineering Company.

Perovskite film fabrication

$Cs_{0.05}FA_{0.85}MA_{0.1}PbI_3$ perovskite precursor (1.75 mM) was fully dissolved in mixed solvents of DMF and DMSO (4:1, v/v), with a FAI/MAI/CsI molar ratio of 0.85:0.1:0.05. Furthermore, 13 mol% MACl and 5% excess PbI_2 were added in the solution to improve the film morphology. The perovskite precursor was then stirred until dissolved at room temperature, and subsequently filtered through a 0.22 μm polytetrafluoroethylene membrane before use. Perovskite precursor solution (50 μl) was spin-coated on substrate at 1,000 r.p.m. for 10 s followed by 5000 r.p.m. for 30 s. Chlorobenzene (200 μl) was dropped onto the substrate 10 s before the end of the spinning, resulting in the formation of dark brown films that were then annealed on a hot plate at 100 °C for 30 min.

Solar cell fabrication

Pre-patterned FTO-coated glass was used as the substrate. The substrates were sequentially sonicated in detergent (Decon 90), deionized water, acetone and IPA for 20 min each. The washed FTO glasses were then dried in an 80 °C oven for over 12 h. The substrates were then exposed to UV–ozone treatment for 20 min before use and transferred into a nitrogen-filled glovebox maintaining less than 0.1 ppm of O_2 and H_2O. Next, 1 ml of CbzNaph (1.5 mM) or CbzNaph:JJ24 (in a molar ratio of 2:1, host CbzNaph concentration = 1.5 mM) in anhydrous ethanol solution was uniformly spread on the substrates, followed by spinning the films at 3,000 r.p.m. for 30 s. Then the CbzNaph and CbzNaph:JJ24 films were annealed at 100 °C for 10 min then washed by dropping 200 μl of ethanol while spinning the film at 3,000 r.p.m., followed by annealing at 100 °C for 10 min for CbzNaph SAM, but annealing the CbzNaph:JJ24 sample at 160 °C for 15 min for to cross-link the CbzNaph:JJ24. Perovskite solution was then deposited on the SAM-modified substrate as detailed above. After the perovskite film was cooled to room temperature, 50 μl of $EDAI_2$ solution (0.25 mg ml^{-1} in IPA) was dynamically drop-cast onto the perovskite film at 5,000 r.p.m. and annealed at 100 °C for 5 min. Devices based on CbzNaph and cross-linked CbzNaph:JJ24 films were then transferred to a thermal evaporator. Then C_{60} (25 nm) was deposited with a rate of 0.3 Å s^{-1} in a high-vacuum chamber ($<2 \times 10^{-6}$ torr). Next, a 20 nm SnO_x layer was deposited using atomic layer deposition (ALD) in the GEMStar XT Bench-Top ALD system (Arradiance). Tetrakis(dimethylamido)tin(IV) (TDMASn) and H_2O were used as tin and oxygen precursors, respectively. A silver electrode (100 nm) was thermally evaporated in the high-vacuum chamber through a metal shadow mask with designed aperture area. Finally, a 10 nm Al_2O_3 layer was deposited on the device with ALD system to serve as an encapsulated layer. Trimethylaluminium and H_2O were used as aluminium and oxygen precursors, respectively. The following encapsulation processes were completed in a nitrogen-filled glovebox. The cell was encapsulated in two glass sheets with edge sealing via polyisobutylene rubber.

Solar cell characterization

The current-voltage (I–V) characteristics of solar cells were measured at room temperature using a Keithley 2400 Source Meter under simulated sunlight from a solar simulator (SS-F5, EnliTech). The devices were measured at a scanning speed of 0.01 V per step with dwell time of 100 ms under the forward and reverse directions. A National-Renewable-Energy-Laboratory-calibrated silicon solar cell (with a KG-2 filter) was used to obtain AM 1.5 G (100 mW cm^{-2}) solar simulator light intensity. The perovskite solar cells were covered with a shading mask with an aperture area of 0.0712 cm^2 to ensure the accuracy of the current density from J–V curves. The EQE curves were obtained using an EQE measurement system (QE-R, EnliTech). The stable power output of solar cells was measured by tracking the current at the voltage of the maximum power point.

Stability tests of solar cells

All of the devices used for stability tests were encapsulated as previously described[42]. The long-term operational stability of the encapsulated devices was tracked by a solar cell stability testing system (91PVKSolar Technology Company) equipped with multichannel and automatic data-collecting elements. The encapsulated solar cells were placed in a chamber without an ultraviolet filter and tracked at their MPPs under continuous 1-sun-equivalent illumination (from a white light-emitting diode lamp) in ambient air with an relative humidity of 70–80%. During MPP tracking, the chamber atmosphere was heated to 85 °C by a temperature-control system. The thermal cycling test was implemented with temperature cycling between −40 °C (10 min dwell) and 85 °C (10 min dwell) in dark, at a ramp rate of 100 °C per hour. The I–V characteristics of devices were conducted after 20, 50, 100, 200, 300, 400, 500 and 700 cycles.

Other characterizations

^1H NMR and ^{13}C NMR spectra were measured on Bruker AVANCE III 300 MHz, 400 MHz and 600 MHz spectrometers. Photoluminescence spectra were collected using a home-built facility, in which an excitation laser (450 nm) was introduced to the sample through a fibre, and the photoluminescence spectra were detected using a detector connected to an Ocean Optics USB2000 spectrometer. X-ray photon spectroscopy was performed on a Thermo Fisher ESCALAB XI + X-ray photoelectron spectrometer. Non-monochromatic He I was used as ultraviolet light with an energy of 21.21 eV. The top-viewing morphology of the thin-film samples and cross-sectional profile of the tandem cell were conducted by SEM (QUATTRO S). Thin-film XRD characterizations were conducted on a D2 Phaser instrument with Cu Kα (wavelength of 1.5418 Å) radiation. Time-resolved photoluminescence was recorded using an FLS980 spectrofluorometer (Edinburgh) with a 485 nm pulsed excitation laser (EPL-485; repetition rate of 2.5 kHz to 20 MHz; peak power of 115.19–222.22 mW; beam size of 1.5 mm^2). Photo-induced force microscopy results were acquired using a VistaScope microscope from Molecular Vista. All PiFM experiments were excited by a pulsed quantum cascade laser with a gap-free narrowband tuneable wavenumber of 760–1,950 cm^{-1} (Block Engineering) and 1,995–2,310 cm^{-1} (Daylight Solutions). In this work the PiFM experiment was operated at the sideband excitation with the laser-frequency modulated at $f_m = f_1 - f_0$, where f_0 is the first mechanical eigenmode resonances of the cantilever that is used for PiF signal detection, whereas f_1 denotes the second ones recorded for the AFM topography of the sample. The probe is an Pt-coated tip with a resonant frequency of ~350 kHz (PPP-NCHPt-MB, Nanosensors). X-ray reflectivity was performed at BL02U2 beamline at Shanghai Synchrotron Radiation Facility. The energy of the X-ray used was 9.1 keV (λ = 0.1363 nm), A Huber six-circle diffractometer combined with a scintillation detector was used to ensure the speed and accuracy of our experiments. The film thicknesses were extracted using the X-Ray Calc software[43].

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data generated or analysed during this study are included in the published article and its Supplementary Information.

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Author contributions W.J., S.W. and A.J. conceived the original idea. W.J. refined the experimental protocol, and designed and synthesized JJ24 under the supervision of A.J. The perovskite films and devices were fabricated by G.Q., S.W. and J.Z. Data analysis was performed by W.J. and G.Q. under the guidance of S.W., J.Z. and A.J. Theoretical calculations were performed by L.C. and T.W. under the supervision of Q.J., X.H. and X.C., and C-T.W. assisted with characterization and device fabrication. F.R.L. and C.Y. provided helpful discussions. W.J. wrote the paper, which was revised by G.Q., S.W., J.Z. and A.J. All authors discussed the results and commented on the paper. A.J. supervised the project.

Competing interests The authors declare no competing interests.

Additional information
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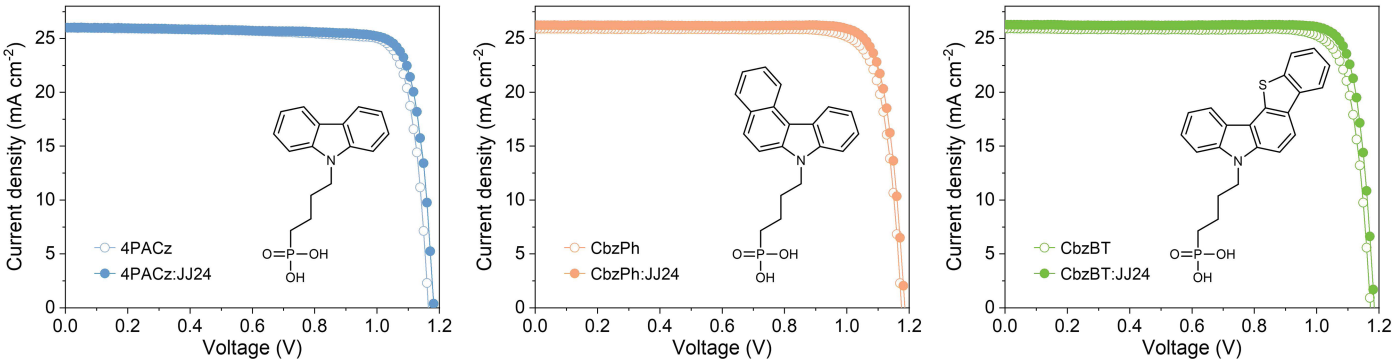
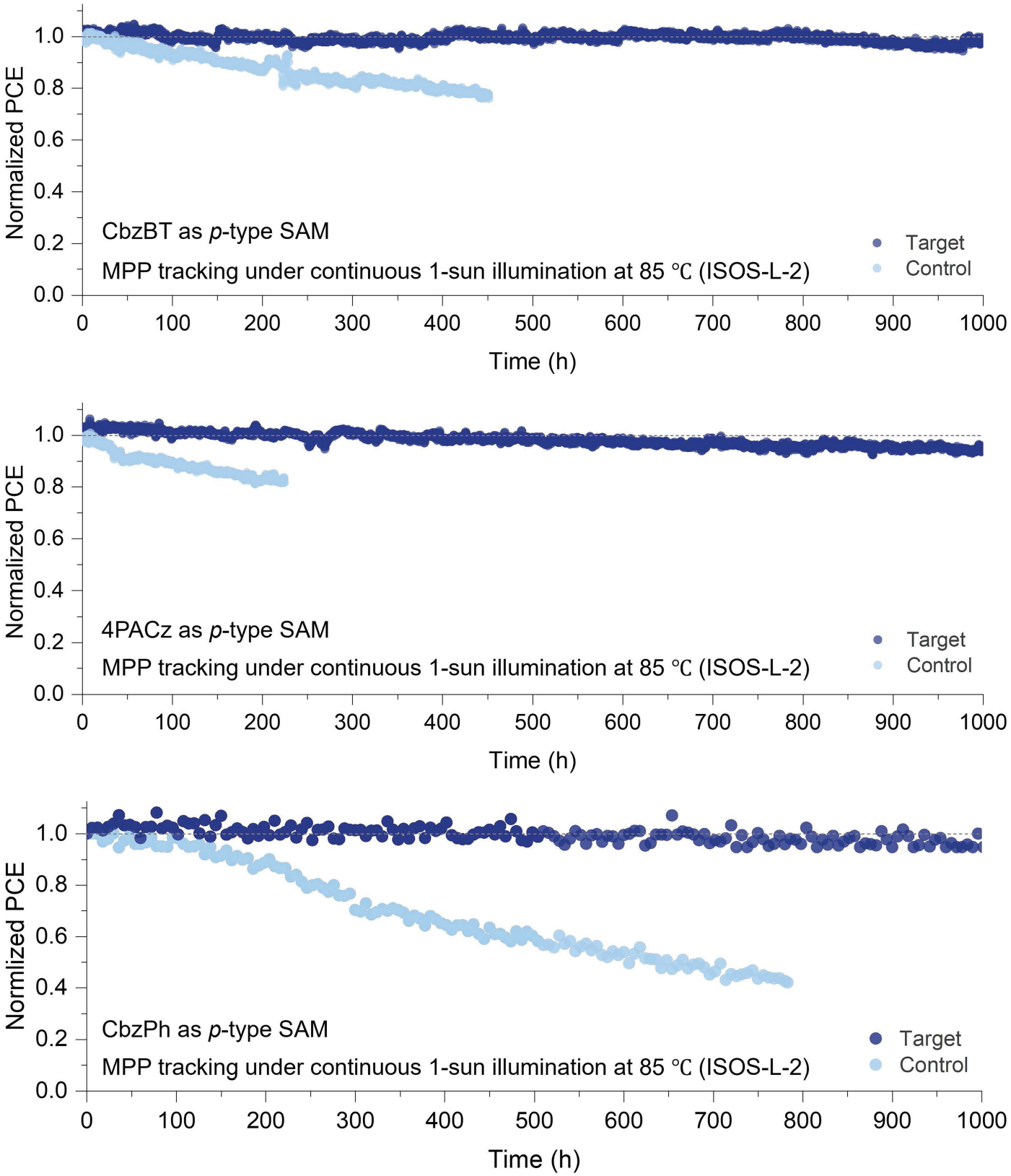
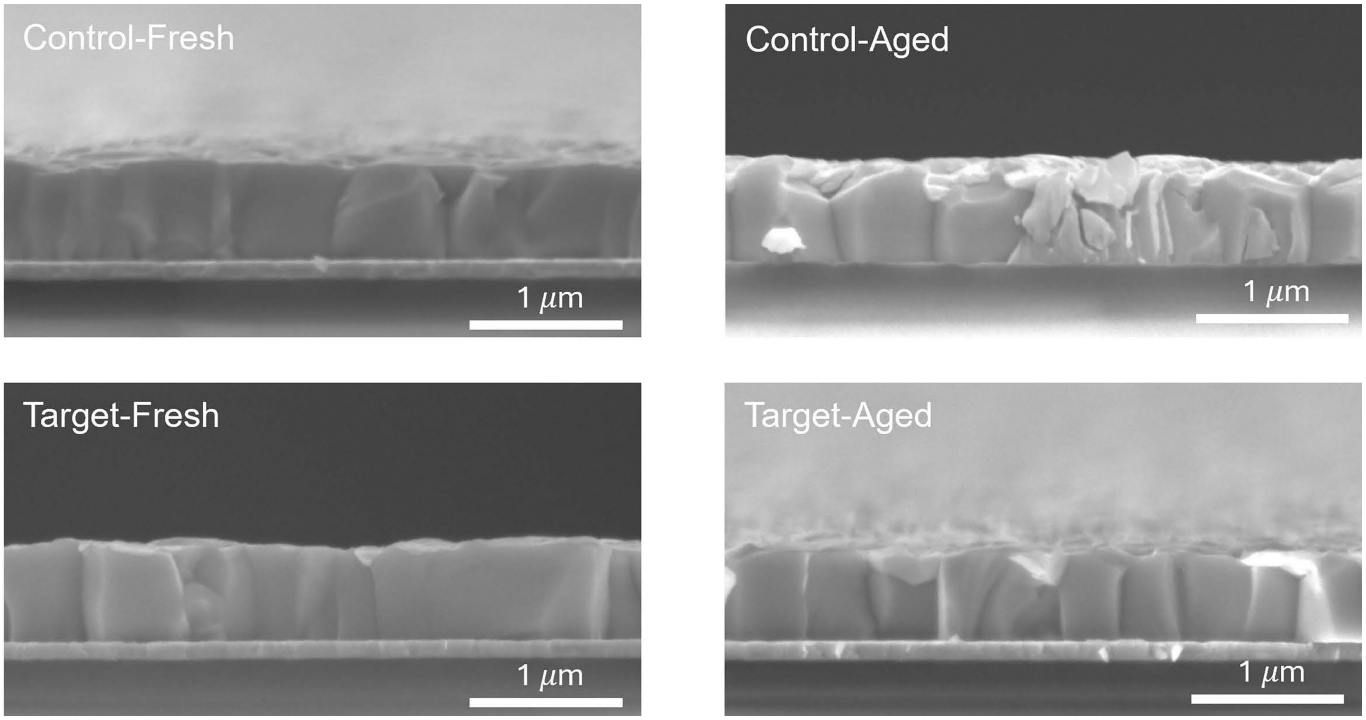


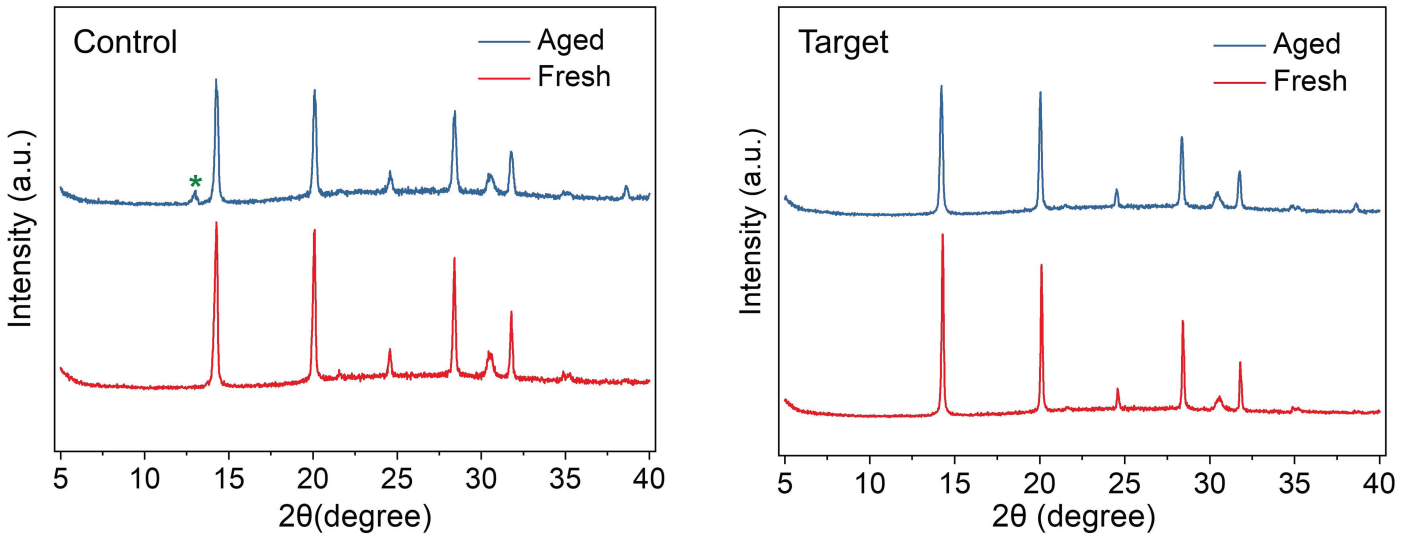
Extended Data Fig. 1 | *J-V* curves of other *p*-type SAMs toughened with JJ24. *J-V* characteristics of the champion PSCs based on 4PACz, CbzPh and CbzBT with and without JJ24 cross-linking.



Extended Data Fig. 2 | Long-term operational stability of devices based other *p*-type SAMs toughened with JJ24. Long-term operational stability of the encapsulated PSCs based on different *p*-type SAM molecules at MPP tracking under continuous 1-sun equivalent illumination at 85 °C in ambient air with RH of 70–80%, complied with the ISOS-L-2 protocol.



Extended Data Fig. 3 | Cross-sectional SEM images of perovskite films. Cross-sectional SEM images of control and target perovskite films before and after ageing (under continuous illumination at 85 °C for 100 h).



Extended Data Fig. 4 | XRD patterns of perovskite films. XRD patterns of control and target perovskite films before and after ageing (under continuous illumination at 85 °C for 100 h).

Extended Data Table 1 | Detailed photovoltaic parameters of other *p*-type SAMs toughened with JJ24

SAMs	V_{OC} (V)	FF (%)	J_{SC} (mA/cm^2)	PCE (%)
4PACz	1.164	83.68	25.98	25.30
4PACz:JJ24	1.182	83.92	26.02	25.81
CbzPh	1.175	83.31	25.92	25.37
CbzPh:JJ24	1.186	84.87	26.23	26.40
CbzBT	1.172	83.82	25.93	25.47
CbzBT:JJ24	1.185	86.07	26.28	26.80

Detailed photovoltaic parameters of the champion PSCs based on 4PACz, CbzPh and CbzBT with and without JJ24 cross-linking.

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Solar Cells Reporting Summary

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Experimental design

Please check the following details are reported in the manuscript, and provide a brief description or explanation where applicable.

1. Dimensions

Area of the tested solar cells	☒ Yes	0.1 cm2 and 1.2 cm2 for small-area and large-area devices, respectively.
	☐ No	*Explain why this information is not reported/not relevant.*
Method used to determine the device area	☒ Yes	The device areas are defined by the metal shading masks during electrode deposition.
	☐ No	*Explain why this information is not reported/not relevant.*

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction	☒ Yes	The J-V plots in both forward and backward direction are provided in Fig. 3c and Supplementary Fig. S21.
	☐ No	
Voltage scan conditions	☒ Yes	From 1.20 V to -0.1 V (backward scan) or -0.1 V to 1.2 V (forward scan) with steps of 0.01 V
	☐ No	*Explain why this information is not reported/not relevant.*
Test environment	☒ Yes	The devices were tested at room temperature in nitrogen-filled glovebox.
	☐ No	*Explain why this information is not reported/not relevant.*
Protocol for preconditioning of the device before its characterization	☐ Yes	*Provide a description of the protocol.*
	☒ No	No preconditioning protocol.
Stability of the J-V characteristic	☒ Yes	Stability of the J-V characteristic were obtained by tracking the maximum power point of cell, as provided in Supplementary Fig. S22.
	☐ No	*Explain why this information is not reported/not relevant.*

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization	☒ Yes	No unusual behavior was found.
	☐ No	*Explain why this information is not reported/not relevant.*
Related experimental data	☒ Yes	See Fig. 3c and Supplementary Fig. S21.
	☐ No	*Explain why this information is not reported/not relevant.*

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)	☒ Yes	see Supplementary Fig. S23 and Methods section for details.
	☐ No	*Explain why this information is not reported/not relevant.*
A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☒ Yes	See main text. The integrated current density of the device (25.43 mA cm-2) matched well with that measured by J-V curve (26.25 mA cm-2) (Less than 5%).
	☐ No	*Explain why this information is not reported/not relevant.*

For tandem solar cells, the bias illumination and bias voltage used for each subcell

☐ Yes
☒ No

Provide a description of the measurement conditions.

Not relevant.

5. Calibration

Light source and reference cell or sensor used for the characterization

☒ Yes
☐ No

See Methods section.

Explain why this information is not reported/not relevant.

Confirmation that the reference cell was calibrated and certified

☒ Yes
☐ No

See Methods section.

Explain why this information is not reported/not relevant.

Calculation of spectral mismatch between the reference cell and the devices under test

☒ Yes
☐ No

The spectrum used for measurement matches well with the reference silicon cell. No spectral mismatch calculations have been performed in our laboratory, but third-party certification has been made to calculate spectral mismatch.

Explain why this information is not reported/not relevant.

6. Mask/aperture

Size of the mask/aperture used during testing

☒ Yes
☐ No

0.0712 cm2 and 1.015 cm2 for small-area and large-area devices, respectively.

Explain why this information is not reported/not relevant.

Variation of the measured short-circuit current density with the mask/aperture area

☐ Yes
☒ No

Report the difference in the short-circuit current density values measured with the mask and aperture area.

For the cells tested in our lab, all cells were measured using one identical mask.

7. Performance certification

Identity of the independent certification laboratory that confirmed the photovoltaic performance

☒ Yes
☐ No

One small-area cell was send out for certification by third-party institution.

Explain why this information is not reported/not relevant.

A copy of any certificate(s)

☒ Yes
☐ No

See Supplementary Fig. 24.

Explain why this information is not reported/not relevant.

8. Statistics

Number of solar cells tested

☒ Yes
☐ No

10 solar cells.

Explain why this information is not reported/not relevant.

Statistical analysis of the device performance

☒ Yes
☐ No

See Fig. 3b, Supplementary Fig. S20, and Supplementary Table S3.

Explain why this information is not reported/not relevant.

9. Long-term stability analysis

Type of analysis, bias conditions and environmental conditions

☒ Yes
☐ No

Long-term device stability of the encapsulated cells under MPPT at 85 ℃ and efficiency tracking of the encapsulated cells subjected to repetitive thermal cycling between -40 ℃ and 85 ℃. See Figs. 3d-3e, Extended Data Fig. 2, Supplementary Fig. S31, and Methods section.

Explain why this information is not reported/not relevant.